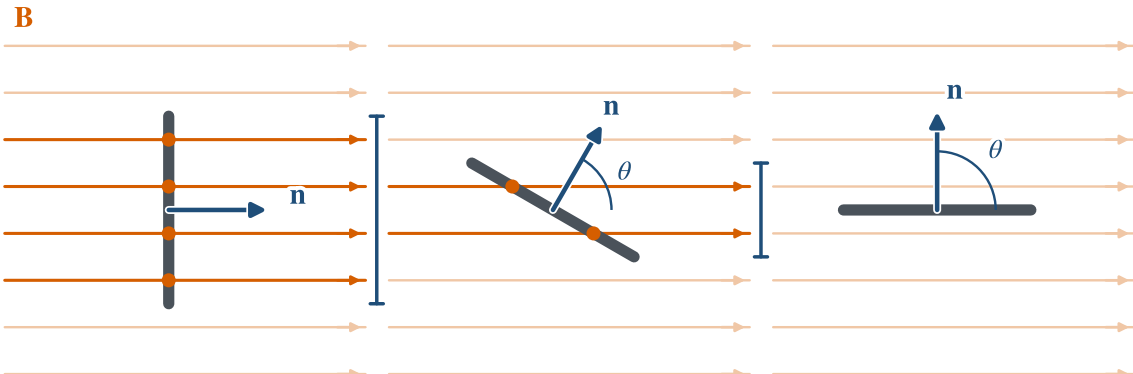
$\theta = 0$: $\Phi_B = BA_c$ $\theta = 60^\circ$: $\Phi_B = BA_c/2$ $\theta = 90^\circ$: $\Phi_B = 0$



bar: projection of the surface onto a plane normal to B, length $\propto A_c \cos\theta$; dots: lines crossing it